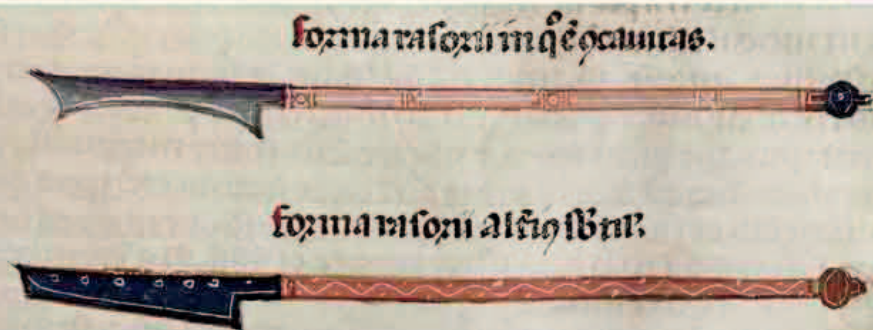


200

THE APPROXIMATE
NUMBER OF **NEW
SURGICAL
INSTRUMENTS**
INTRODUCED
BY AL-ZAHRAWI



A 14th-century manuscript shows two innovative surgical instruments introduced by Spanish-Arab physician al-Zahrawi (Albucasis).

THE MODERN

NUMERALS for expressing the **decimal system** first appeared in Europe in 976 in a treatise written by the monk Vigila in the northern Spanish convent of Albelda (although he included only the symbols for 1 to 9 and not zero). This number system, now known as Hindu-Arabic, had originated in the numerical notation of the Brahmi script used in India in the mid-3rd century BCE, which then spread westward after the Arabs came in contact with India in the early 8th century CE.

Although **mechanical armillary spheres** and **mechanical escapements for clocks** had been devised in China by Zhang Heng in the 2nd century and Yi Xing in the 8th century, a superior version was constructed by



Abacus

This is a modern example of an abacus, a counting device that appeared in Mesopotamia around 2700 BCE. It was introduced to medieval Europe by Gerbert in around 990 CE.

the astronomer **Zhang Sixun** in 979. It was powered by a waterwheel with scoops that deposited the liquid into a clepsydra (a device that measures time by the flow of liquid through a small hole) as it turned around, which in turn regulated the measurement of the hours. In order to avoid the liquid freezing in winter—a problem that had afflicted earlier such clocks—Yi Xing had substituted **mercury** for water. Zhang Sixun's improved version made one complete revolution each day, with each quarter-hour and hour sounded

out by mechanical jacks that emerged to strike bells and drums, or to display the time on a tablet. The clock also showed the **position of the Sun, Moon, and five planets on a celestial globe**, and was said to be so advanced that after Sixun's death no one could keep it in working order.

In 984, Persian mathematician **Ibn Sahl** (c.940–1000) wrote *On the Burning Instruments*, a treatise in which he examined the bending of light by lenses and curved mirrors. He was the first to express a **geometric theory of refraction**. He suggested that the amount of light that is deflected when it enters another medium (such as glass) varies, depending on the refractive index (see 1621–24) of the substance.

Christian monastic scholar **Gerbert** (c.943–1003), who became Pope in 999, was one of the **first Western European mathematicians** of the Middle Ages. He sought to recover mathematical and astronomical

treatises by ancient scholars such as Boethius, studied the work of Islamic mathematicians, and **introduced the abacus to Europe**, giving instructions for its use in multiplication and division.

The greatest Arab surgeon of medieval times was **Abu al-Qasim al-Zahrawi** (c.936–1013),

also known as Albucasis. He was court surgeon to al-Hakam, the Umayyad Caliph of Cordoba. His *Kitab al-Tasrif* (*The Method of Medicine*)—containing detailed descriptions of **human anatomy** and the **pathology of diseases**—became the main textbook for medieval European physicians.

“...HE WHO DEVOTES HIMSELF TO **SURGERY** MUST BE VERSED IN... **ANATOMY.**”

Al-Zahrawi (Albucasis), in *Kitab al-Tasrif*, c.990

Babylonian	Ancient Egyptian	Ancient Greek	Ancient Roman	Ancient Chinese	Mayan	Modern Hindu-Arabic
𐎶		α	I	一	•	1
𐎵		β	II	二	••	2
𐎴		γ	III	三	•••	3
𐎳		δ	IV	四	••••	4
𐎲		ϵ	V	五	—	5
𐎱		ς	VI	六	—•	6
𐎰		ζ	VII	七	—••	7
𐎯		η	VIII	八	—•••	8
𐎮		θ	IX	九	—••••	9
𐎭		ι	X	十	—•••••	10

DEVELOPMENT OF NUMBERS

Many early number systems, such as the Egyptian, were additive—the value of the number symbol did not depend on its position; to make 20, the symbol for 10 would be written twice. Around 2000 BCE, the Babylonians began to use a partly positional system—where the order of magnitude depends on the position in which the symbol appears. Positional systems using 10 as the base developed in India, and gradually evolved into the modern Hindu-Arabic numerals.

c.932 Jewish physician and philosopher Isaac ben Solomon describes malarial fever

c.950–60 Abu al-Khazini produces tables for making astrolabe disks

c.950–79 Ibn al-Jazzar writes treatises on geriatrics and sleep disorders

969 Al-Sijzi produces work on the trisection of an angle

c.980 Abu ibn al-Abbas (Haly Abbas) writes *Al-Maliki*, one of main Arabic medical textbooks

979 Zhang Sixun builds a mechanical clock powered by mercury

979–984 Ibn Sahl proposes the law of refraction of light

c.990 Gerbert introduces the abacus to medieval Europe

994 Persian astronomer al-Khujandi discovers that Earth's axial tilt is decreasing